

Enhancing the Goldilocks Algorithm for Optimism Season 7

Executive Summary

The Goldilocks algorithm is a key component of Optimism's Retro-Funding Evaluation, designed to reward onchain builders (projects) that demonstrate steady and balanced contributions across multiple metrics, with a strong focus on retention. However, as Optimism Season 7 shifts its priority to driving Total Value Locked (TVL), the current weight configuration reveals a significant limitation: it favors established projects while overlooking "rising stars"—emerging projects with shorter track records but substantial growth potential and impact. This exclusion risks undervaluing innovative contributors critical to achieving Season 7's TVL objectives. This report examines these shortcomings, proposes refined metric and variant weights, and demonstrates how these adjustments better align with Optimism's goal of rewarding impact while preserving the algorithm's foundational strengths.

1. Introduction

Optimism's Retro-Funding initiative aims to promote growth, adoption, and sustainability within the Superchain ecosystem. The Goldilocks algorithm supports this mission by assessing projects through a combination of metrics and variants, historically emphasizing retention to encourage consistent contributions. With Season 7's strategic focus on driving Total Value Locked (TVL), however, the algorithm's current design—featuring equal metric weights and a retention-heavy structure—falls short. A major limitation is its inability to adequately recognize rising stars, whose rapid growth and potential impact are vital to ecosystem expansion. This report details the deficiencies of the existing setup and proposes targeted adjustments to better align the algorithm with Season 7's TVL-driven priorities, while maintaining its commitment to balanced performance evaluation.

2. Shortcomings of the Current Weight Configuration

The current Goldilocks algorithm employs the following weights:

Current Metric Weights

Metric	Weight	Description
<code>trace_count</code>	0.25	Raw transaction activity
<code>monthly_average_tvl</code>	0.25	Average TVL over a month
<code>amortized_gas_fee</code>	0.25	Sustained economic activity
<code>monthly_active_farcaster_users</code>	0.25	Farcaster platform engagement

Current Variant Weights

Variant	Weight	Description
<code>adoption</code>	0.20	Current period value
<code>growth</code>	0.20	Change from previous period
<code>retention</code>	0.60	Minimum of current and previous

2.1 Equal Metric Weights Undermine TVL Focus and Exclude Rising Stars

- *Issue:* All metrics are weighted equally at 0.25, despite total value locked (TVL) being Season 7's primary target. Metrics like `trace_count` and `monthly_active_farcaster_users`, while informative, do not directly contribute to TVL and dilute the focus on value-centric projects. Furthermore, this equal weighting overlooks the unique strengths of rising stars, which often demonstrate strong growth metrics rather than established performance.
- *Impact:* Established projects with stable but stagnant metrics may outscore rising stars with significant growth potential, misaligning rewards with Season 7's goal of ecosystem expansion through TVL growth.

2.2 Vulnerability to Artificial Activity

- *Issue:* The lack of bot-filtered metrics (e.g., `transaction_count_bot_filtered`) allows unverified or artificial activity to influence scores.
- *Impact:* Projects with inflated metrics due to bot activity could receive undeserved rewards, compromising fairness and the recognition of genuine value creation.

2.3 Platform Bias Towards Farcaster

- *Issue:* Relying exclusively on `monthly_active_farcaster_users` restricts community assessment to a single platform.
- *Impact:* Projects with robust engagement on alternative platforms (e.g., X, Discord) are undervalued, reducing inclusivity and potentially excluding rising stars that thrive on diverse channels.

3. Proposed Adjustments

The proposed changes recalibrate the algorithm by structuring the metric weights into three key categories—*TVL Drivers*, *Transactions*, and *User Metrics*—following a 50-30-20 weighting approach. This framework reflects Retro-Funding’s [methodology](#) of evaluating onchain impact through financial commitment (TVL), transaction activity, and user adoption, while emphasizing growth to spotlight rising stars.

Proposed Metric Weights

Metric	Current Weight	Proposed Weight	Rationale
<code>trace_count</code>	0.25	0.20	Reduced weight from 0.25 to 0.20 to balance the introduction of bot-filtered transaction counts, ensuring a more accurate measure of genuine activity.
<code>monthly_average_tvl</code>	0.25	0.30	Increased weight from 0.25 to 0.30 to prioritize TVL as a key driver of ecosystem value, aligning with Season 7’s focus on liquidity and financial commitment.
<code>transaction_count_bot_filtered</code>	0.00	0.10	Introduced with a 0.10 weight to capture authentic transaction activity, mitigating the risk of inflated metrics and enhancing fairness in transaction evaluation.
<code>amortized_gas_fee</code>	0.25	0.20	Reduced weight from 0.25 to 0.20 to accommodate new user metrics while still valuing economic activity as a measure of network utilization.

monthly_active_farcaster_users	0.25	0.10	Decreased weight from 0.25 to 0.10 to balance with the introduction of monthly active addresses, focusing on a more diversified user engagement assessment.
monthly_active_addresses	0.00	0.10	Introduced with a 0.10 weight to better capture on-chain user adoption, providing a more comprehensive view of a project's real impact on developer teams and users.

Proposed Variant Weights

Variant	Current Weight	Proposed Weight	Rationale
adoption	0.20	0.10	Reduced weight from 0.20 to 0.10 to shift focus toward growth, ensuring emerging projects with strong momentum are not overshadowed by established ones.
growth	0.20	0.30	Increased weight from 0.20 to 0.30 to spotlight rising stars by rewarding projects with significant growth, aligning with Season 7's goal of fostering ecosystem expansion.
retention	0.60	0.60	Maintained weight at 0.60 to ensure stability and consistency remain key factors, balancing the increased focus on growth while valuing sustained performance.

Understanding the 50-30-20 Metric Weight Categories

- **TVL Drivers (50%):**
 - monthly_average_tvl: 0.30
 - amortized_gas_fee: 0.20
- **Transaction Drivers (30%):**
 - trace_count: 0.20
 - transaction_count_bot_filtered: 0.10
- **User Metrics (20%):**
 - monthly_active_farcaster_users: 0.10
 - monthly_active_addresses: 0.10

This 50-30-20 split prioritizes financial commitment (TVL) as the primary driver, supported by transaction activity and user adoption metrics. The increased growth weight of 0.30 in

the variant weights ensures that rising stars—projects demonstrating strong upward momentum—are adequately recognized.

4. Rationale for Proposed Changes

These adjustments address previous shortcomings, align the algorithm with Season 7's TVL-focused objectives, and ensure rising stars are rewarded through a clear, impact-driven structure.

4.1 Prioritizing TVL Drivers (50% Weight)

- *Change:* `monthly_average_tvl` increases from 0.25 to 0.30.
- *Reason:* As the primary measure of ecosystem value, TVL is elevated to anchor the 50% TVL Drivers category. This shift ensures projects contributing to Season 7's core goal are prioritized, particularly rising stars with growing TVL, while `amortized_gas_fee` (0.20) complements it by capturing sustained economic activity.

4.2 Incentivizing Growth to Recognize Rising Stars

- *Change:* `growth` increases from 0.20 to 0.30; `adoption` decreases from 0.20 to 0.10.
- *Reason:* Elevating the `growth` variant rewards projects with upward trajectories, a defining trait of rising stars critical for ecosystem expansion. The strong `retention` weight (0.60) maintains stability, while the reduced `adoption` weight shifts focus from current scale to future potential.

4.3 Ensuring Authentic Transaction Activity (30% Weight)

- *Change:* Introduces `transaction_count_bot_filtered` at 0.10; reduces `trace_count` from 0.25 to 0.20.
- *Reason:* The Transaction Drivers category (30%) now balances raw activity (`trace_count`) with authentic engagement (`transaction_count_bot_filtered`). This ensures rewards reflect genuine usage, which is especially important for rising stars that may face early-stage metric inflation.

4.4 Broadening User Metrics (20% Weight)

- *Change:* Reduces `monthly_active_farcaster_users` from 0.25 to 0.10; adds `monthly_active_addresses` at 0.10.
- *Reason:* Diversifying community metrics mitigates platform bias and captures broader user adoption. This fosters inclusivity, benefiting rising stars active across multiple platforms beyond Farcaster.

4.5 Refining Existing Metrics

- *Change:* `amortized_gas_fee` decreases from 0.25 to 0.20; `trace_count` decreases from 0.25 to 0.20.
 - *Reason:* These reductions free up weight for more strategic metrics like `monthly_average_tvl` and `transaction_count_bot_filtered`. This refines the algorithm to focus on meaningful growth indicators, ensuring rising stars are evaluated on impactful contributions rather than less relevant signals.
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5. Conclusion

The proposed adjustments to the Goldilocks algorithm offer a strategic overhaul that directly addresses its existing limitations while aligning with the core objectives of Optimism Season 7. These refinements ensure the algorithm not only drives Total Value Locked (TVL) but also fosters a fair, inclusive, and resilient Superchain ecosystem. Here's how these changes deliver impactful improvements:

- *Prioritizing TVL:* By assigning a higher weight to `monthly_average_tvl`, the algorithm shifts its focus to reward projects that significantly contribute to TVL, a key metric for Optimism Season 7's success.
- *Balancing Retention and Growth:* The updated framework enhances the `growth` variant to incentivize expansion, while still valuing retention, ensuring both rising stars and established projects are fairly rewarded for their contributions.
- *Enhancing Authenticity:* Introducing bot-filtered metrics mitigates the risk of artificial activity, strengthening the integrity and reliability of the evaluation process.
- *Broadening Inclusivity:* Expanding community metrics beyond platform-specific biases (e.g., Farcaster) promotes diverse engagement, making the algorithm more equitable and representative of the broader ecosystem.

These adjustments align the Goldilocks algorithm with Optimism Season 7's mission to drive TVL, while preserving its foundational commitment to rewarding consistent and

balanced project performance. By adopting this refined framework, Optimism empowers both established contributors and emerging innovators, laying the groundwork for sustained growth, fairness, and profitability of the Superchain ecosystem.